Band-1 Hadamard Order 2164 via Miyamoto — Paley-II H_540 + Paley Conference C_542

H_540 Paley-II PASS; H_2164 Miyamoto PASS; SPK gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-14_Band1_Hadamard_next_gap_after_2036_triage`; `2026-09-14_Band1_Hadamard_508_to_`
Lab. `/workspace/experiments/band1/hadamard_2164_miyamoto/`.
Cutoff. 2026-09-14 (~evening ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 540 and 2164 only** (verified max $|HH^\top - nI| = 0$). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 2164 is a **Miyamoto bonus** (not an SPK fill). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed H_{508}, H_{2004}, H_{2036}, H_{528}, and H_{2116}. Among remaining Miyamoto-eligible standard-closure misses flagged after the 2036 triage, this increment constructs **order 2164**: $2164 = 4{\cdot}541$ with 541 prime and $541 \equiv 1 \pmod 4$. On box: H_{540} via Paley type II ($q = 269$, $269 \equiv 1 \pmod 4$), symmetric Paley conference C_{542} over prime 541, and $H_{2164} = \text{Miyamoto}(H_{540}, C_{542})$. All verify $\max |HH^\top - nI| = 0$. Unit tests **17/17 PASS**. SPK next gap remains **2052**; $2164 \notin$ SPK closure (Miyamoto bonus). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 2164; Miyamoto; Paley type II; Paley conference; constructive gap 2052; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in $\{\pm 1\}$ satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{540} constructed and verified (Paley type II, $q = 269$). 2. C_{542} constructed and verified (Paley conference, $q = 541$): $CC^\top = 541I$, zero diagonal. 3. H_{2164} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 4. SPK next constructive gap remains **2052** (OPEN). Order 2164 does **not** enter SPK.

2. Arithmetic and eligibility

Quantity	Value
Target order	$2164 = 4 \cdot 541$
$q = 541$	prime, $541 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{540}
Needed conference	order $542 = q + 1$
H_{540} path	Paley II: $q' = 269$ prime $\equiv 1 \pmod 4$; order $2(269 + 1) = 540$
Conference path	Symmetric Paley over prime 541

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{540}	Paley type II, $q = 269$	0	0.0981	**PASS**
C_{542}	Paley conference, $q = 541$	0	0.2033	**PASS**
H_{2164}	Miyamoto(H_{540}, C_{542})	0	4.2948	**PASS**

Total wall (incl. unit tests + SPK refresh) $\approx$ 4.8454 s. Compressed ± 1 matrices saved as `H540_pm1.npz`, `H2164_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible $\leq 2000 \cup$ on-box $\{508, 2004, 2036, 528, 2116\} \cup$ verified H_{540} (Paley-II seed) $\cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4000). **Miyamoto is not folded into SPK.**

Quantity	Value
Known count	877
2164 in SPK	**False** (Miyamoto bonus)
2116 in SPK	False (prior Miyamoto bonus)
Next standard gap	**2052** (OPEN)
Missing first (sample)	2052, 2060, 2076, ...

5. Unit tests (17/17)

Test	Result
`H20_paley_I_control`	PASS
`H4_sylvester`	PASS
`paley_II_q5_H12`	PASS
`paley_II_q13_H28`	PASS
`miyamoto_H20_from_H4_conf6`	PASS
`q541_is_prime`	PASS
`q541_equiv_1_mod_4`	PASS
`n2164_equals_4_times_541`	PASS
`q269_is_prime`	PASS
`q269_equiv_1_mod_4`	PASS
`H540_order_formula`	PASS
`C542_order_formula`	PASS
`reject_all_ones_H8`	PASS
`H540_paley_II_eligible`	PASS
`conference_542_shape_expected`	PASS
`conference_6_paley_control`	PASS
`miyamoto_eligible_2164_arith`	PASS

6. Non-claims

- Not the full Hadamard conjecture.

- Not Clay-closed / finished-ToE.
- Not a fill of standard gap **2052**.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims.
- GATE-04 collection **LOCKED**; no invented J.
- `clay_closed = false`.

7. Next optional

Miyamoto **2228** (need H_{556}) / **2276** (need H_{568}); OR method-attack standard gap **2052**; OR Track B privacy / Object 1 meter await.

8. Reproducibility

```
cd /workspace/experiments/band1/hadamard_2164_miyamoto
python3 verify_hadamard_2164.py
# emits results.json, SUMMARY.txt, literature_ledger.json,
# H540_pm1.npz, H2164_pm1.npz
```

Artifacts. `results.json`, `SUMMARY.txt`, `literature_ledger.json`, `H540_pm1.npz`, `H2164_pm1.npz`, this paper.